Optimization & Outlook: Hotel Reservation Data Analysis

Strategies for Increasing Direct Traffic and Operational Efficiency (2015-2017)

Strategic Data Review prepared for Hotel Management.

Executive Summary: From Insight to Action

The Goal

Increase direct website visits and improve operational efficiency across Resort and City hotel properties.

The Diagnosis

- **Retention Gap**: Direct channels drive volume but fail to retain guests (13.5k new vs. 934 repeat).
- **Cancellation Risk**: City Hotel faces a critical 41.7% cancellation rate, driven by long lead times (>90 days).
- **Operational Friction**: A 12.36% room mismatch rate indicates allocation inefficiencies.

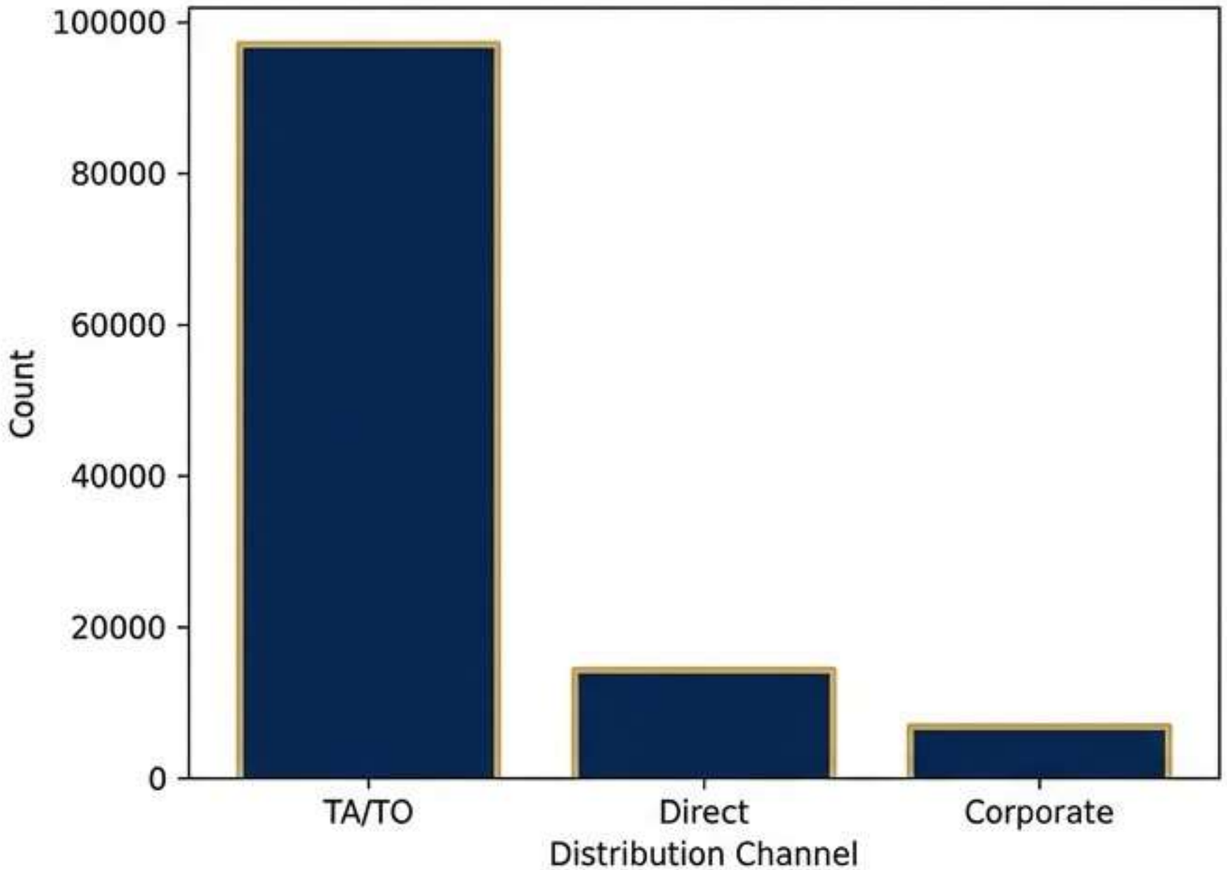
The Forecast

Prophet Time-Series model significantly outperforms LightGBM (RMSE 1670 vs. 6361), proving distinct seasonal demand patterns.

The Strategic Fix

- **Marketing**: Launch loyalty programs for 'Direct' channel retention.
- **Operations**: Enforce non-refundable deposits for bookings >90 days out.
- **Staffing**: Align rosters with weekly and yearly demand curves.

Direct and TA/TO Channels Drive Volume, but Efficiency Varies



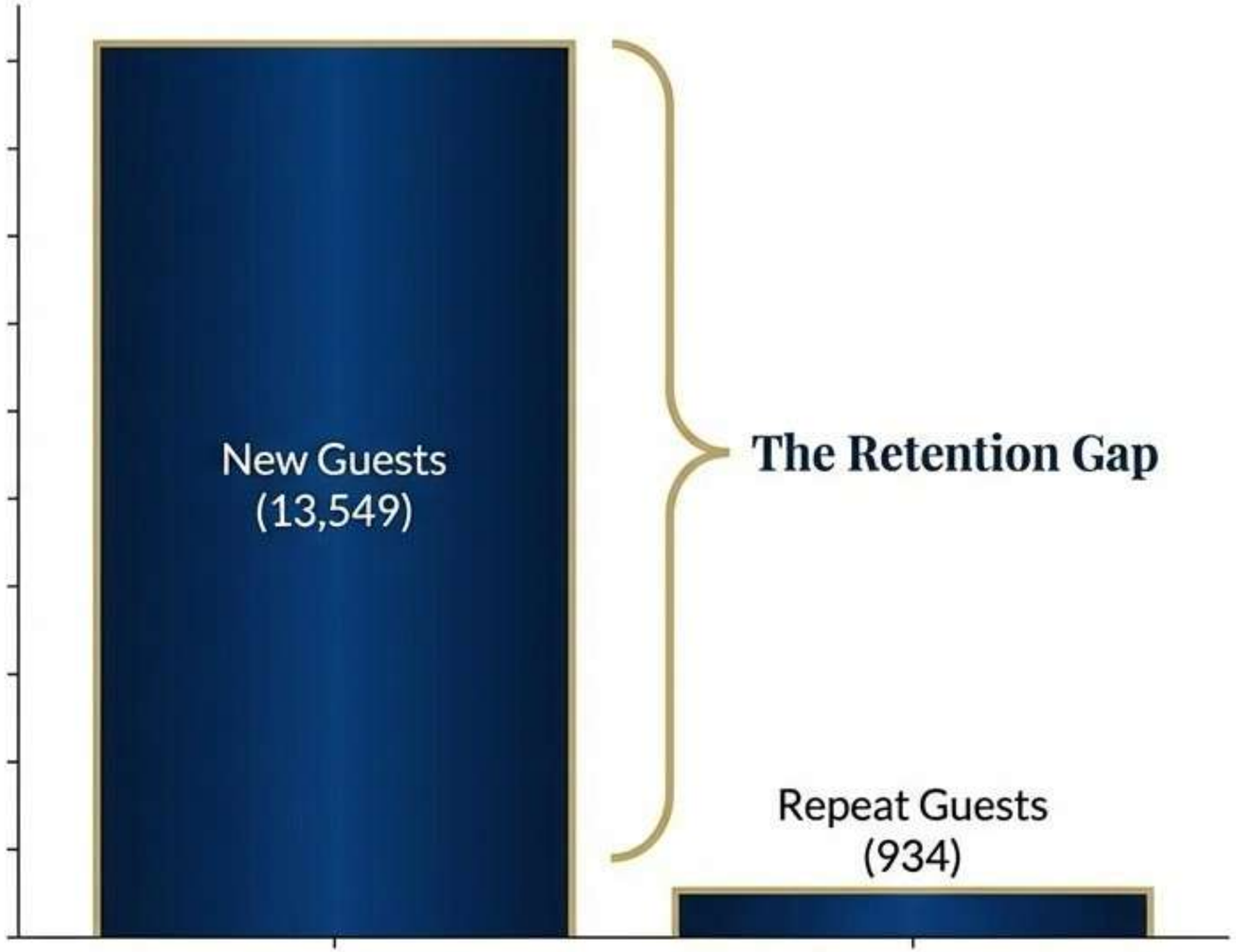
Key Metric:

Direct Channel Efficiency: 82.4% successful booking rate.

TA/TO Efficiency: 58.9% successful booking rate.

Insight: While Travel Agents/Tour Operators bring the most volume, the Direct channel is significantly more efficient at converting interest into stays.

The Direct Channel Attracts New Guests but Struggles to Retain Them

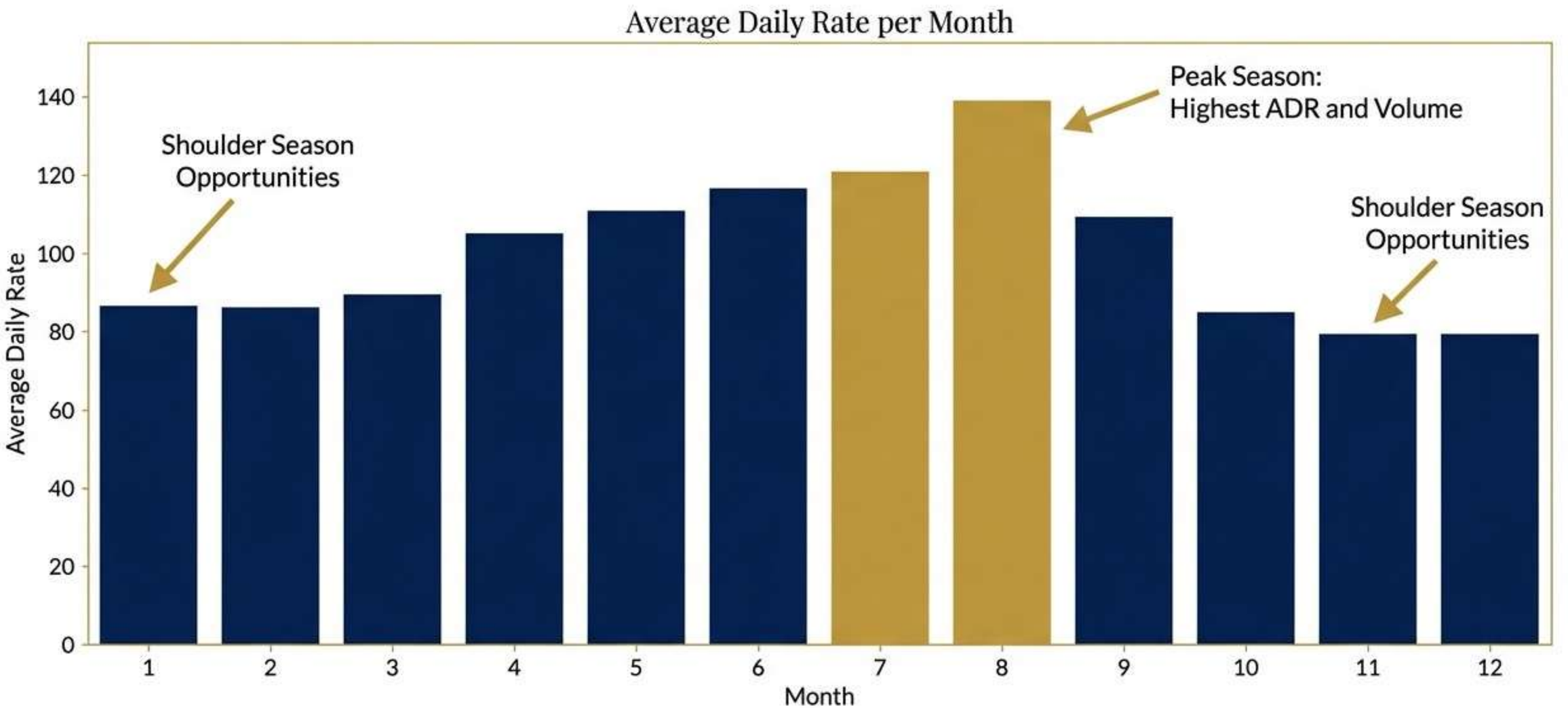


There is a massive disparity between acquisition and retention.

We are successfully driving traffic to the website, but failing to incentivize a second visit.

Targeting these 13,500+ “one-time” guests is the lowest-hanging fruit for revenue growth.

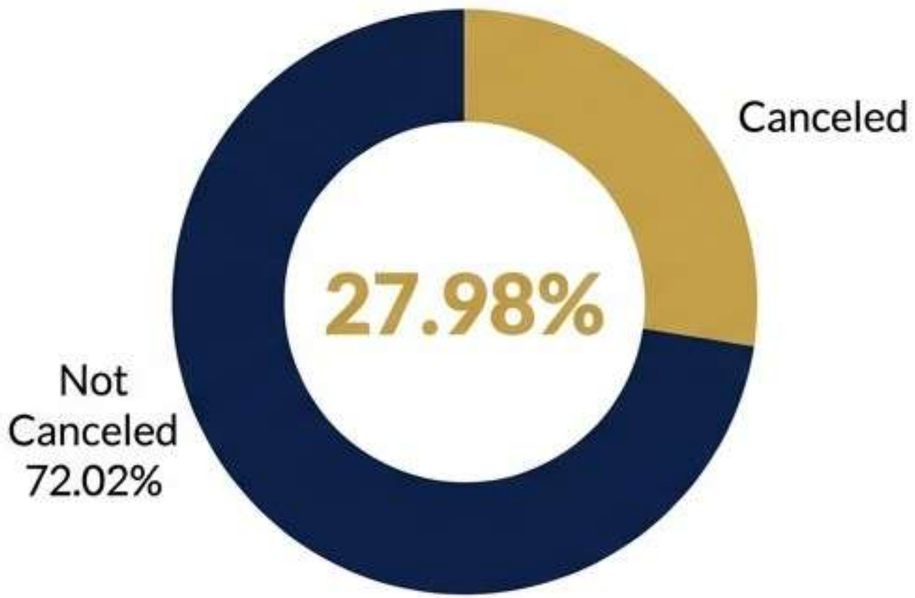
July and August Represent Peak Pricing Power and Demand



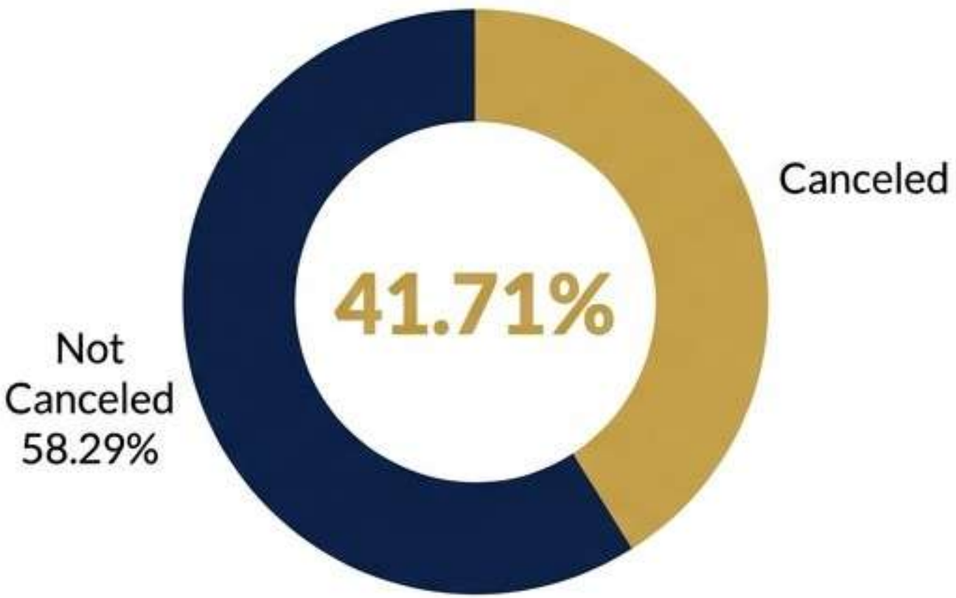
Strategy: Maximize yield during Summer peaks; use value-add promotions to stimulate demand in November.

City Hotel Operations Are Severely Impacted by High Cancellations

Resort Hotel

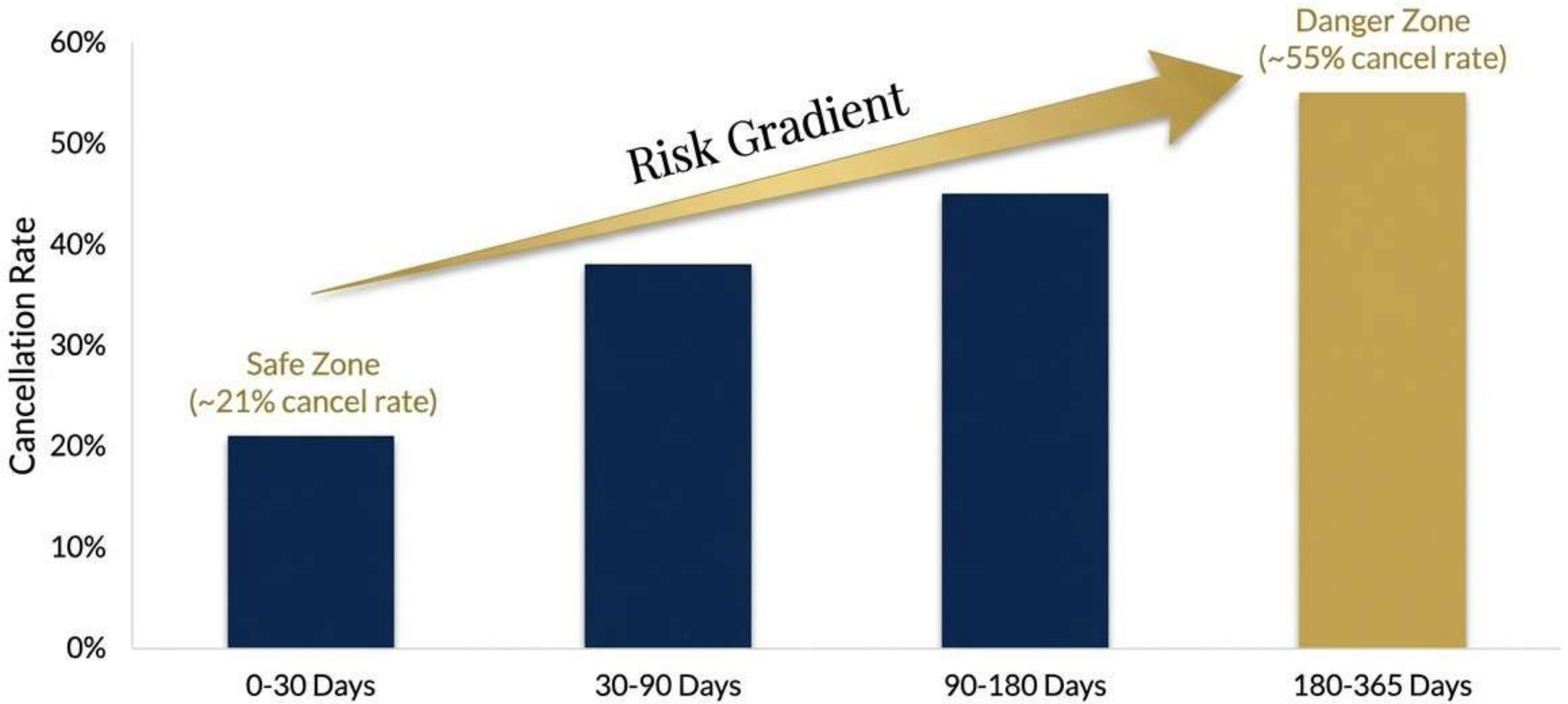


City Hotel



4 out of 10 bookings at the City property do not result in a stay, causing massive inventory management issues.

Longer Lead Times Drastically Increase Cancellation Risk



Financial Commitment via Non-Refundable Deposits Eliminates Cancellations

THE SILVER BULLET

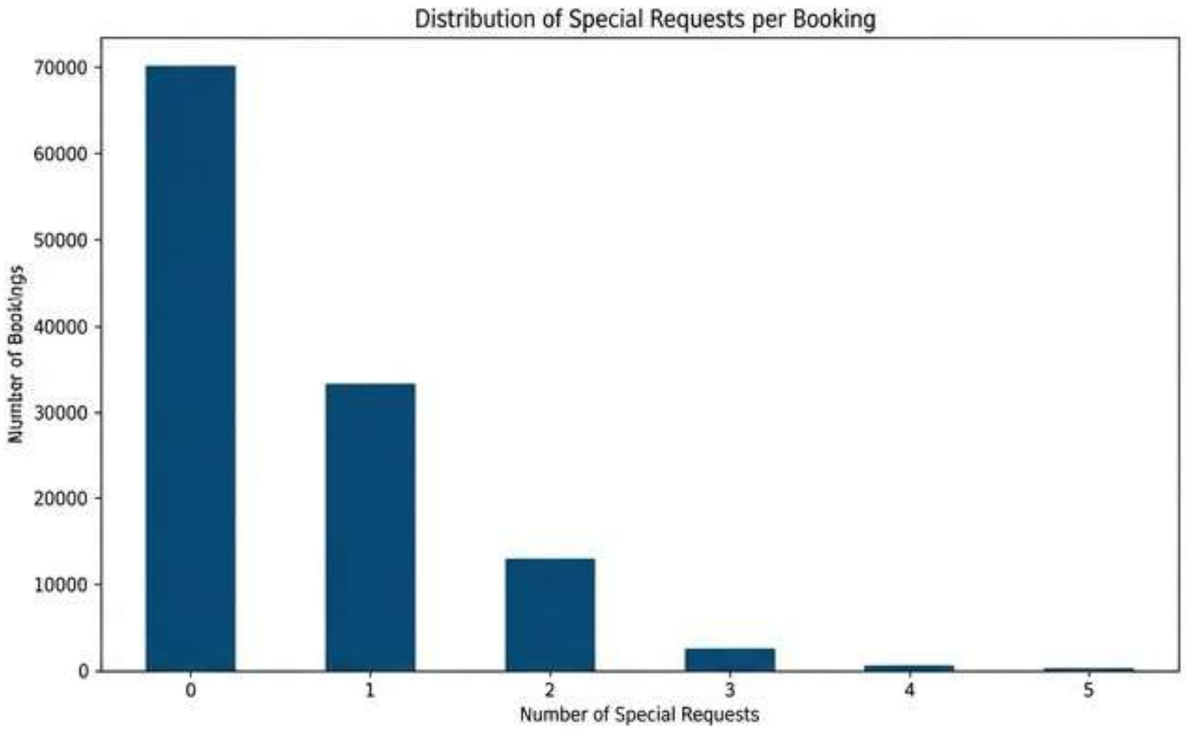
Non-Refundable deposits achieve >99% guest retention.

We must apply this policy to the high-risk >90 day lead time segment.

12% of Guests Do Not Receive Their Reserved Room Type

12.36%

Room Mismatch Rate



The majority of guests (69k) have zero special requests, suggesting the mismatch is an internal allocation error.

Immediate audit of the PMS Room Allocation process is required to protect guest satisfaction.

Leveraging Time-Series AI to Accurately Predict Visitor Volume

Model A: LightGBM

FAILED. RMSE: 6361. Failed to capture seasonal nuances.

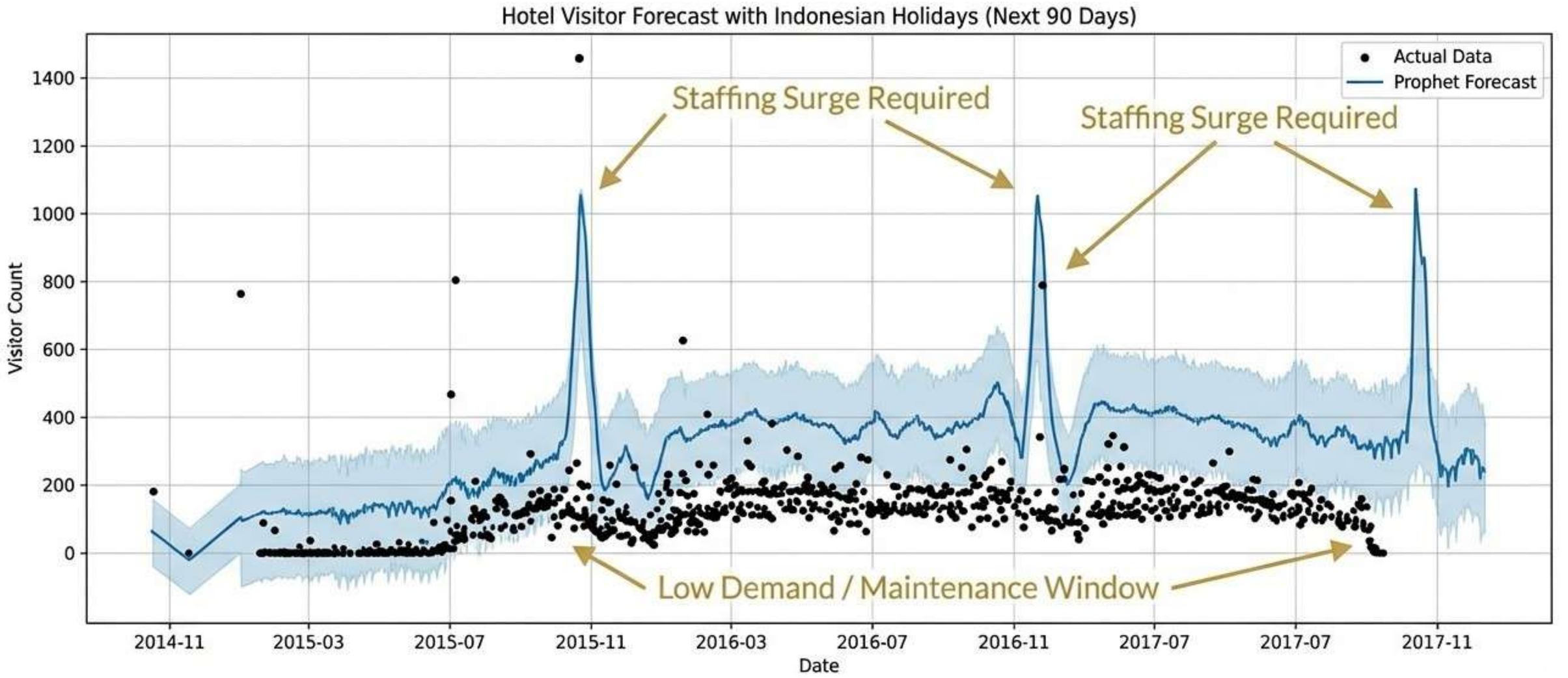
Model B: Prophet

PASSED. RMSE: 1670. Successfully modeled complex seasonal peaks.

We are proceeding with the Prophet model to guide staffing and inventory.

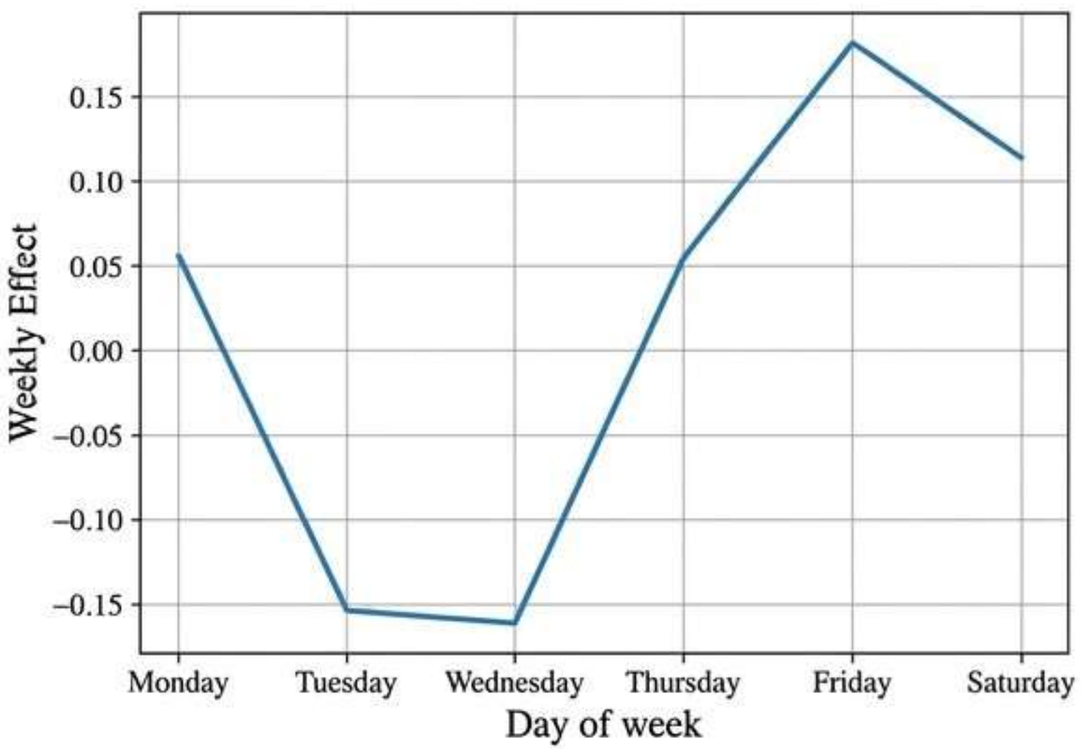
The 90-Day Demand Outlook

Anticipating holiday effects prevents under-staffing during critical surges.



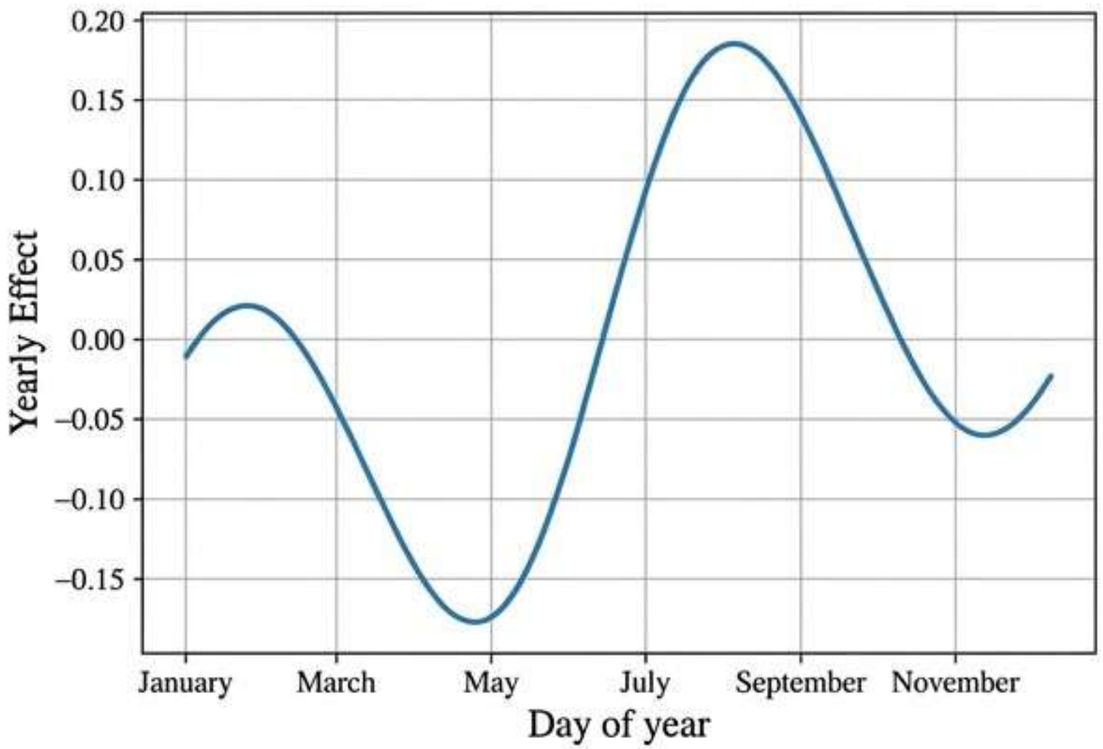
Deconstructing Demand Cycles for Roster Optimization

The Weekly Cycle



Action: Reduce variable staff on Tue/Wed.

The Yearly Cycle



Action: Maximize headcount July-August.

Strategic Recommendations: Marketing & Growth

Fix the Leaky Bucket

Target the ~13k one-time direct bookers with a specific 'Return & Save' loyalty campaign to improve the 6.4% repeat rate.

Dynamic Seasonal Pricing

Aggressively increase ADR in July/August to capture willingness-to-to-pay. Create value-bundles for November to support volume.

Mobile Optimization

Enhance the landing page experience for mobile users to capitalize on the 82% direct conversion efficiency.

Strategic Recommendations: Operations & Efficiency

Risk-Based Deposit Policy

Implement mandatory non-refundable deposits for all bookings made >90 days in advance to mitigate the ~50% cancellation rate in this segment.

Operational Audit

Investigate PMS allocation logic to reduce the 12.36% room mismatch rate which endangers guest satisfaction.

Forecast-Led Staffing

Use the Prophet 90-day rolling forecast to set weekly labor budgets, reducing overtime costs during Tuesday/Wednesday demand troughs.

A Data-Driven Path to Higher Efficiency

- RISKS IDENTIFIED: City Hotel cancellations and Long Lead Time exposure.
- OPPORTUNITIES FOUND: Direct channel retention and Peak pricing power.
- TOOLS DEPLOYED: Prophet Forecast Model for real-time decision making.

Next Steps: Immediate deployment of new deposit policy and loyalty outreach campaign.